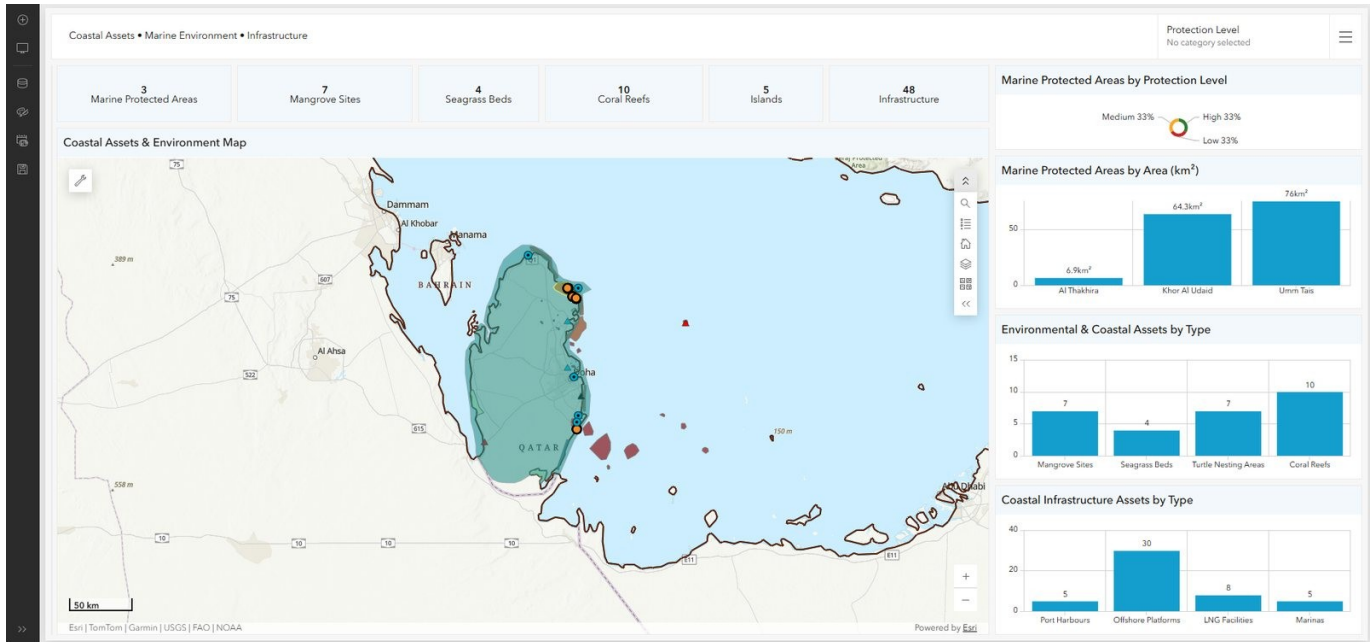


# QATAR COASTAL GIS DASHBOARD

## Personal GIS Portfolio Project - Case Study

ArcGIS Pro | ArcGIS Online | Web Map | ArcGIS Dashboards | Python / ArcPy



*Final interactive Qatar Coastal GIS Dashboard*

**From GIS data preparation and QA/QC to an interactive public web dashboard.**

Note: This is a personal GIS portfolio project created using sample/public data and is not an official government dashboard.

## 1 | Project Objective

The objective was to build an end-to-end coastal GIS solution for Qatar that combines environmental and infrastructure datasets and presents them through an interactive management-style dashboard.

- Develop and organize GIS datasets in a file geodatabase.
- Apply attribute standardization, QA/QC and selected Python/ArcPy automation.
- Publish the data to ArcGIS Online and configure a clear Web Map.
- Build indicators, charts, filters and map interactions in ArcGIS Dashboards.
- Prepare a public, view-only portfolio experience that can be opened without an ArcGIS sign-in.

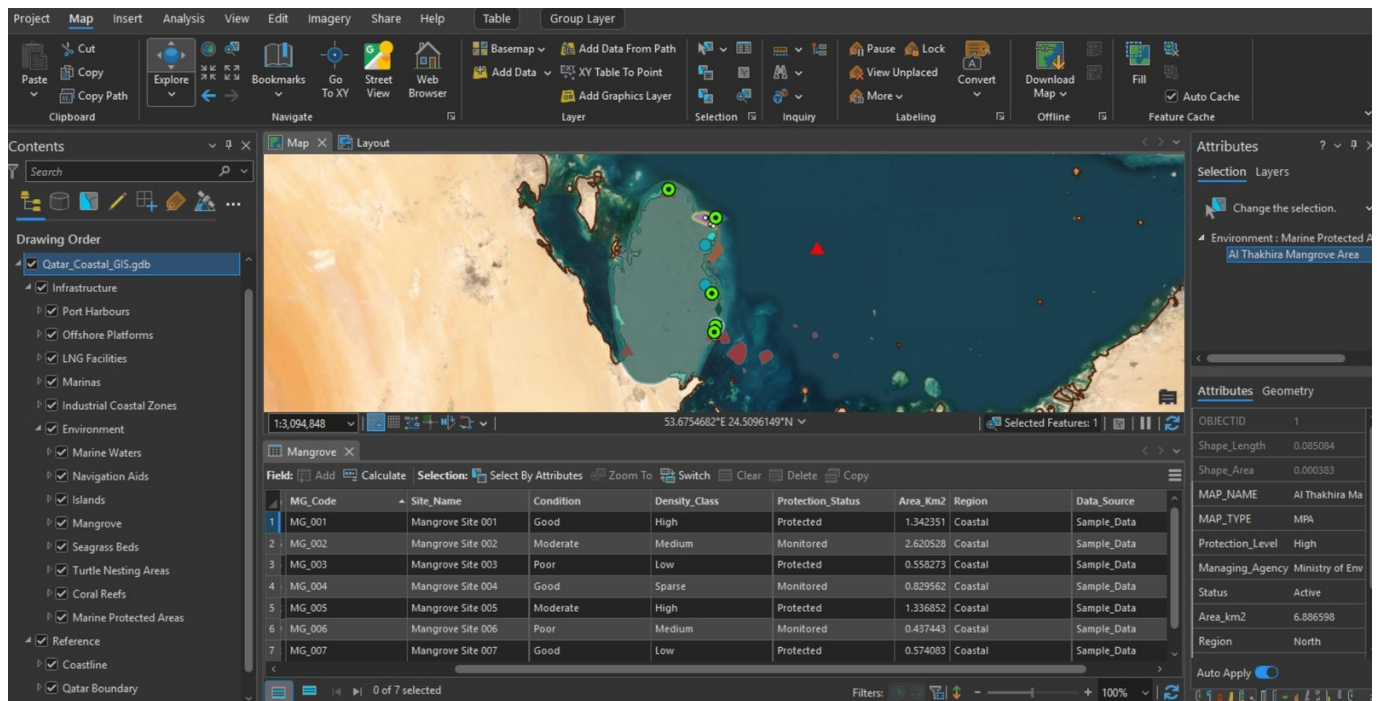
### Project Themes

| Environment                                                                                           | Infrastructure                                                                               | Reference / Context                                             |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Marine Protected Areas<br>Mangrove<br>Seagrass Beds<br>Turtle Nesting Areas<br>Coral Reefs<br>Islands | Port Harbours<br>Offshore Platforms<br>LNG Facilities<br>Marinas<br>Industrial Coastal Zones | Marine Waters<br>Navigation Aids<br>Coastline<br>Qatar Boundary |

### Workflow

ArcGIS Pro → QA/QC + ArcPy → Hosted Feature Layer → Web Map → ArcGIS Dashboard → Public Portfolio

## 2 | Data Development in ArcGIS Pro



*ArcGIS Pro project showing organized layers and attribute review.*

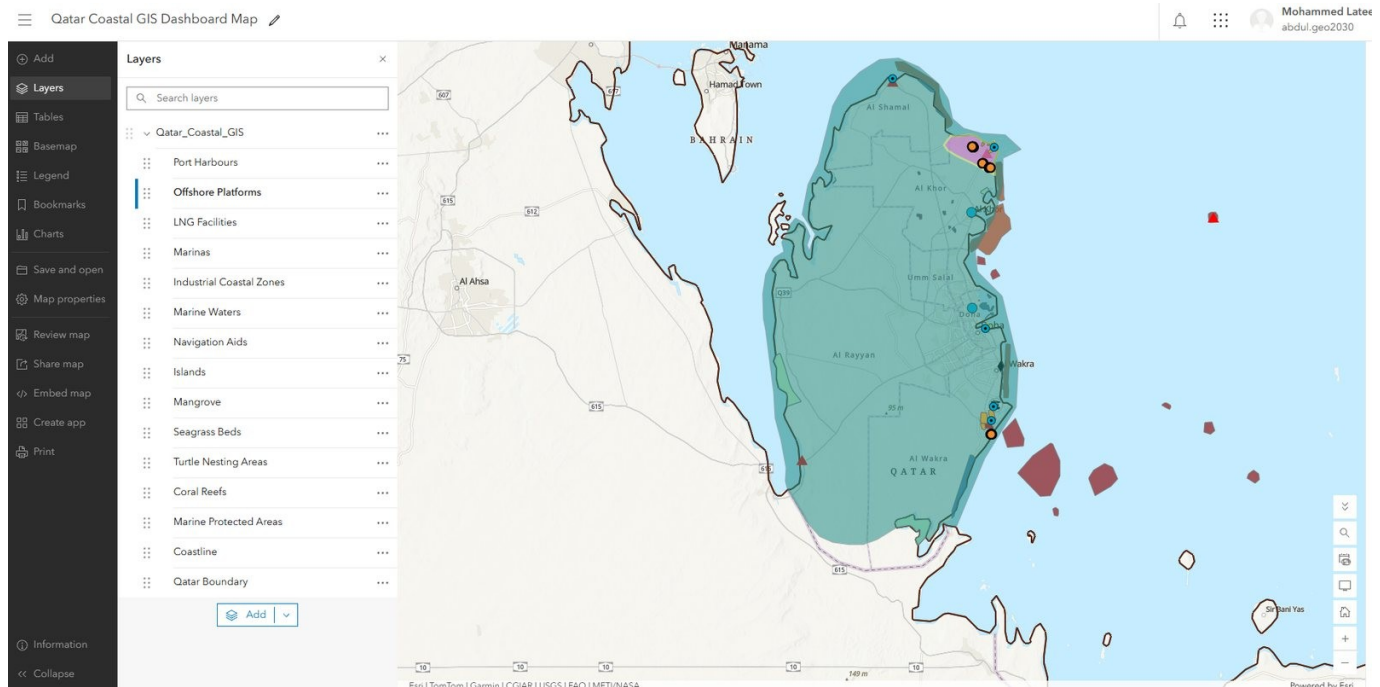
- Designed descriptive fields for codes, names, types, status, region, operator/managing agency, dates and data source.
- Used ArcPy UpdateCursor for repetitive attribute population and sequential codes.
- Calculated polygon areas geodesically in square kilometres.
- Checked field names/types before automation and corrected null or inconsistent values.
- Reviewed feature counts and map symbology before web publishing.

### Example ArcPy Learning

*Sequential IDs: f'SG\_{i:03d}'*

*Geodesic area: geometry.getArea('GEODESIC', 'SQUAREKILOMETERS')*

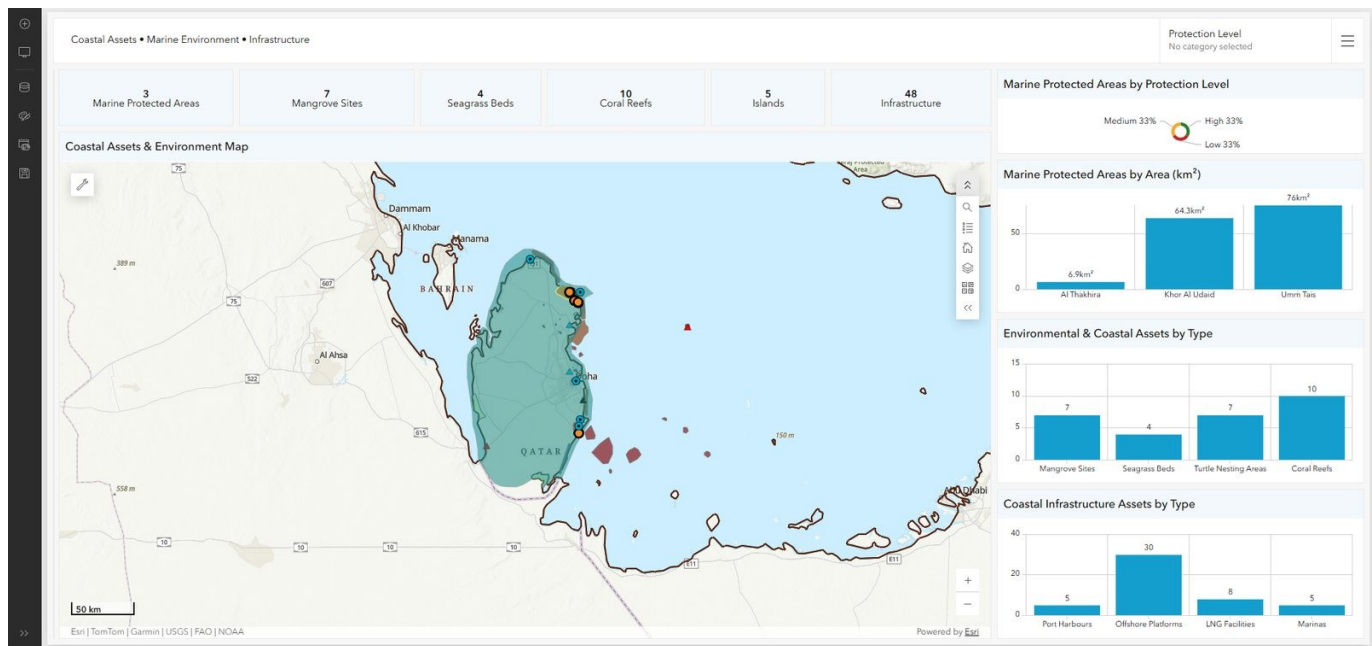
### 3 | Web Map & Cartographic Design



*Web Map after differentiating coastal infrastructure symbols.*

- Configured layer visibility, pop-ups and map legend for dashboard use.
- Reduced Marine Waters transparency to keep the basemap readable.
- Removed the Qatar Boundary fill and retained a clean outline.
- Reduced coastline width so it did not dominate the thematic layers.
- Used distinct symbols for ports, offshore platforms, LNG facilities, marinas and other assets.
- Saved a Qatar-focused Web Map viewpoint so the dashboard Initial View returns to the correct extent.

## 4 | Dashboard Design

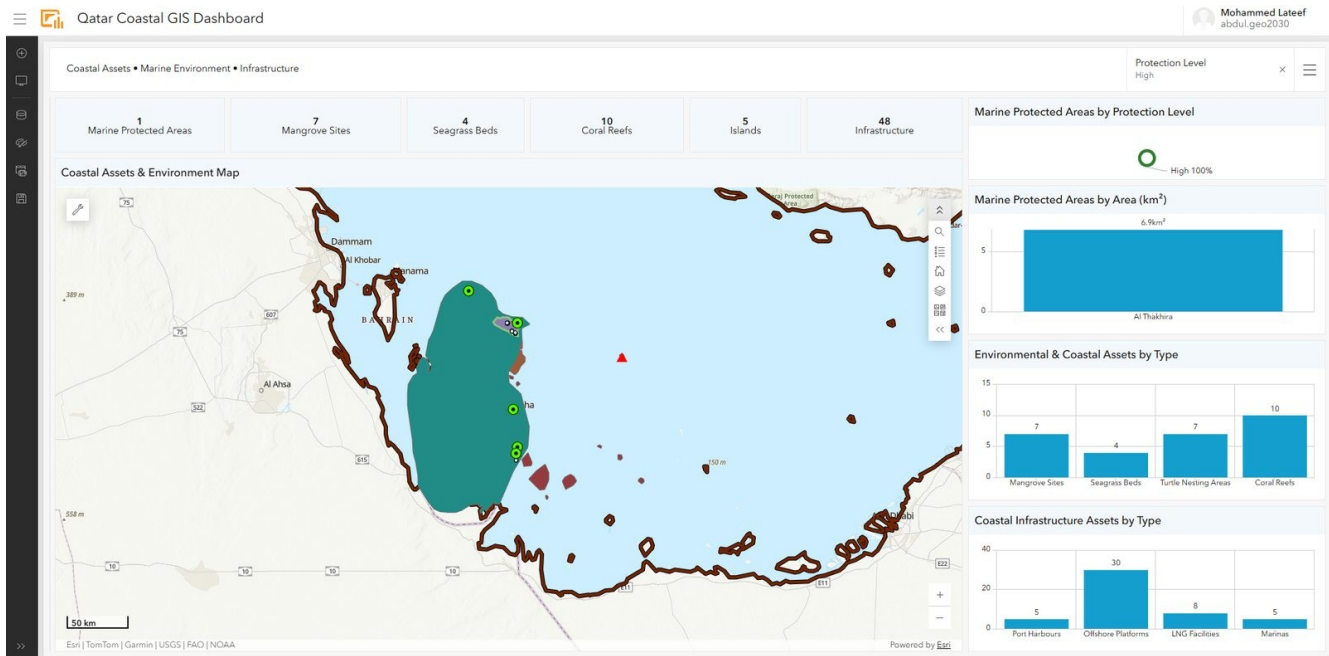


Completed dashboard layout with KPI cards, map, selector and analytical charts.

- Six KPI indicators summarize Marine Protected Areas, Mangrove Sites, Seagrass Beds, Coral Reefs, Islands and Infrastructure.
- Charts show MPA protection levels, MPA area, environmental asset counts and infrastructure asset counts.
- A large central map provides spatial context and interactive exploration.
- Consistent light-blue headers and KPI backgrounds create a simple management-style visual hierarchy.

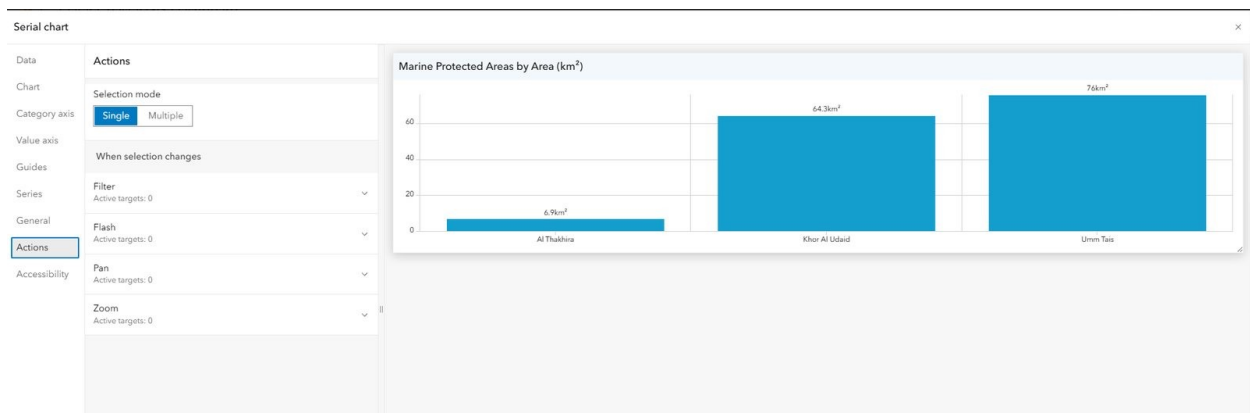
Final headline counts displayed in the completed dashboard: 3 Marine Protected Areas, 7 Mangrove Sites, 4 Seagrass Beds, 10 Coral Reefs, 5 Islands and 48 Infrastructure assets.

## 5 | Interactive Analysis



*Protection Level selector filtering the Marine Protected Area dashboard components.*

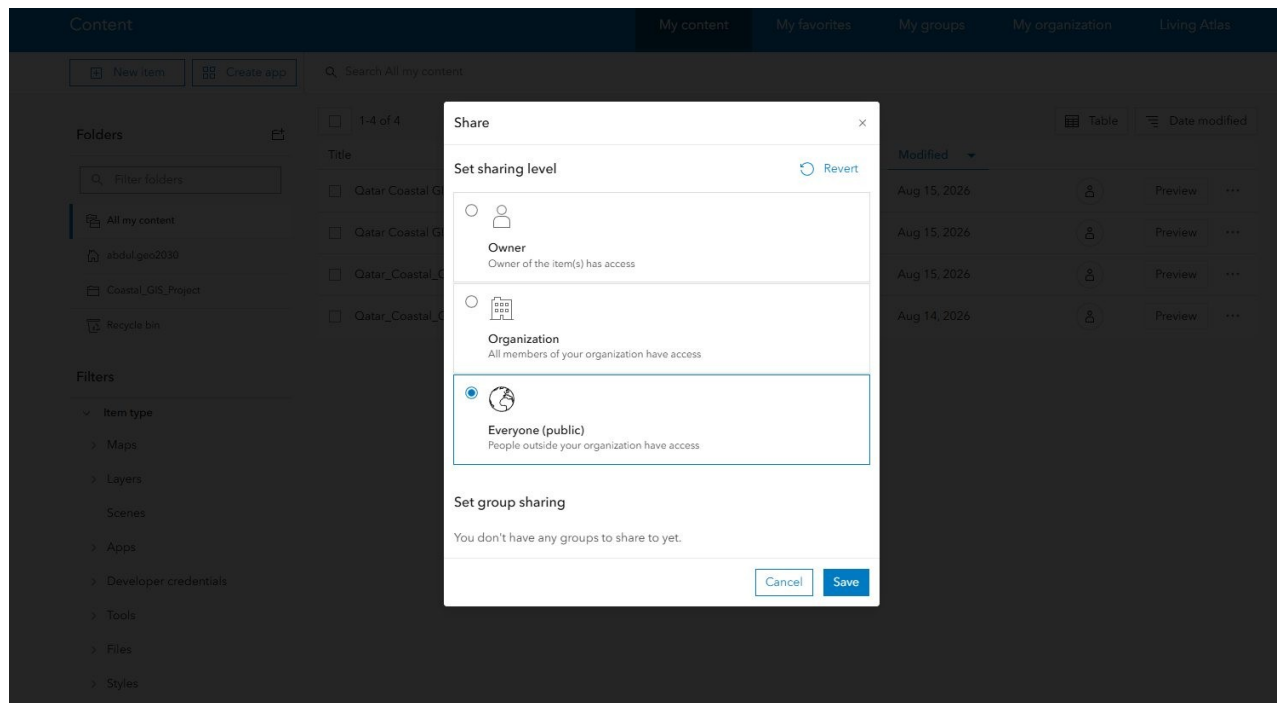
- Protection Level selector uses the Marine Protected Areas Protection\_Level field.
- High, Medium and Low selections filter the MPA KPI, relevant MPA charts and the MPA map layer.
- Unrelated environmental/infrastructure charts remain unchanged because the selector field belongs specifically to the MPA dataset.



*Feature-based MPA Area chart configured with map actions.*

The MPA Area chart uses single feature selection with Zoom and Flash actions, allowing a chart selection to immediately locate and highlight the corresponding protected area on the map.

## 6 | Publishing, Sharing & QA



*Dashboard sharing configured for public portfolio access.*

- Published the GIS content as a hosted feature layer in ArcGIS Online.
- Disabled editing for the public portfolio layer.
- Shared the hosted feature layer, Web Map and Dashboard with Everyone (public).
- Tested the direct dashboard URL in an Incognito/InPrivate browser to confirm anonymous access.
- Verified selectors, chart actions, layer controls, legend and Initial View after publishing.

### Key Troubleshooting Lessons

| Issue                           | Resolution                                                              |
|---------------------------------|-------------------------------------------------------------------------|
| Field not found / type mismatch | Validate the actual schema and field types before running UpdateCursor. |
| Initial View too wide           | Save the desired Qatar viewpoint in the underlying Web Map.             |
| No Zoom/Flash on summary charts | Use feature-based charts for geometry-driven spatial actions.           |
| Public URL asks for sign-in     | Check sharing on every dependency, not only the Dashboard item.         |



## 7 | Skills Demonstrated & Project Result

- GIS data development and geodatabase organization
- Attribute schema design and QA/QC
- Python / ArcPy attribute automation
- Geodesic area calculations
- ArcGIS Online hosted feature publishing
- Web Map configuration and cartographic refinement
- ArcGIS Dashboard indicators, charts and selectors
- Cross-filtering and map actions
- Public sharing, dependency checks and anonymous-access testing
- Portfolio documentation and technical communication

### Project Result

The final deliverable is an interactive Qatar Coastal GIS Dashboard that demonstrates the complete path from desktop GIS data preparation to a public Web GIS application.

#### Interactive Dashboard:

<https://lateefgis2030.maps.arcgis.com/apps/dashboards/1b0ea27771ac4a33bce5cc323e63566e>

**Thank you for viewing this GIS portfolio case study.**

ArcGIS Pro • ArcGIS Online • ArcGIS Dashboards • Python / ArcPy • Web GIS • QA/QC